

TT and LLp represent clinically stable poles. The intermediate forms (BT, BB, BL, LLs) are immunologically unstable and may shift along the spectrum:

- **Upgrading (reversal reaction):** Shift toward stronger immunity (e.g., BT → TT), often accompanied by inflammatory reactions.
- **Downgrading:** Shift toward weaker immunity (e.g., BT → BB → BL), typically asymptomatic but can lead to increased bacterial burden.

Notably, patients with LLs do not downgrade to LLp, and LLp patients generally do not upgrade. The term **LL** is often used to encompass both LLs and LLp.

The pathogenesis of this spectrum is primarily attributed to **type IV hypersensitivity** (delayed-type hypersensitivity, DTH), reflecting the level of **cell-mediated immunity (CMI)** against *M. leprae*.

Classification is based mainly on clinical and histological findings, with bacillary load serving as a secondary criterion.

Terminology Comparison

- **"Tuberculoid"** = TT and BT
- **"Borderline" or "dimorphic"** = BB and BL
- **"Lepromatous"** = LLs and LLp

Acid-fast bacilli are typically **not detectable** in TT and most BT cases, whereas they are **readily found** in BB, BL, LLs, and LLp.

Ridley's classification remains valuable for assessing immune status and guiding management.

Peripheral Nerve Changes

Five common types of peripheral nerve abnormalities occur in leprosy:

1. **Nerve enlargement**, often asymmetric, particularly affecting superficial nerves such as the:
2. Great auricular nerve

3. Ulnar nerve
4. Radial cutaneous nerve
5. Superficial peroneal nerve
6. Sural nerve
7. Posterior tibial nerve
8. **Sensory impairment** within skin lesions.
9. **Nerve trunk palsies**, which may present:
 10. With signs and symptoms of inflammation (e.g., pain, tenderness, swelling),
or
 11. Without overt inflammation (**silent neuropathy**),
Often resulting in both sensory and motor deficits (weakness, muscle atrophy), and, if chronic, joint contractures.
12. **Autonomic dysfunction** in the affected nerve distribution, leading to:
 13. Dry, cracked skin
 14. Anhidrosis (loss of sweating)
 15. Increased risk of injury and ulceration
16. **Neuropathic sequelae**, including:
 17. Chronic ulcers
 18. Osteomyelitis
 19. Painless trauma due to loss of protective sensation
 20. Claw hand, foot drop, and other deformities

Early recognition of nerve involvement is critical to prevent irreversible damage and disability.